

Dhruv Varshney | 23EX10013

Bachelor of Science in Exploration Geophysics (4-Year), Dept. of Geology and Geophysics
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Profile Overview

I am Dhruv Varshney, a motivated second-year undergraduate student pursuing a B.S. in Exploration Geophysics at IIT Kharagpur, with a strong passion for artificial intelligence, machine learning, and data analysis. I am actively developing core skills in Python programming (Pandas, PyTorch, TensorFlow, etc.), building scalable data pipelines, preprocessing intricate datasets, and implementing statistical ideas through hands-on projects.

My areas of interest include:

- Building systematic skills for data interpretation
- Web data extraction and analysis
- Employing basic machine learning models

I am cultivating these skills through directed coursework and experiential experimentation, aiming to solve real-world problems using methodical analytics pipelines. I am seeking a summer internship to apply AI concepts and enhance my technical expertise professionally.

Academic Qualifications

#	Year	Degree/Exam	Institution	Board	CGPA/Percentage
1	2023 - Present	B.S. Exploration Geophysics	IIT Kharagpur	-	8.29/10
2	2023	Senior Secondary Examination	Kendriya Vidyalaya IIT Kharagpur	CBSE New Delhi	91.2%
3	2021	Secondary	Kendriya Vidyalaya IIT Kharagpur	CBSE New Delhi	97.6%

Projects & Internships

Vedvani: Transcribing Spoken Sanskrit Shlokas

May'24

Supervisor: Prof. Pawan Goyal, CSE Department, IIT Kharagpur.

- Scraped a website using BeautifulSoup to gather a dataset of transcripts and corresponding audio files.
- Cleaned and preprocessed the dataset to ensure consistency and accurate alignment between audio and text.
- Trained a Wav2Vec2 model on the prepared dataset to perform speech-to-text transcription for Sanskrit.
- Evaluated model performance using Word Error Rate (WER) and Character Error Rate (CER) metrics, achieving measurable improvements.

Predicting delays in flights

November 2024

OPEN-IIT Competition on Data-Analytics

- Analysed flight schedule data, including departure/arrival times and delay reasons, to develop predictive models for optimizing airline operations.
- Built machine learning models to predict flight delays and reschedule flights, minimising operational disruptions while adhering to constraints like aircraft availability and crew scheduling.
- Evaluated model performance on unseen routes and aircraft, achieving improved generalisation and measurable delay reduction.
- Secured 4th place in a college-level data analytics competition, competing for innovative application of data-driven methodologies.

Achievements

- Secured All India Rank (Gen.) 11603 in JEE Advanced 2023 and secured All India Rank 21317(Gen.) in JEE Mains.
- Secured General Rank 946 in WBJEE (West Bengal Joint Entrance Exam).

Coursework & Skills

- **Programming Languages:** Python, C++, C, HTML, CSS (essential)
- **AI/ML Tools:** TensorFlow, PyTorch, scikit-learn, Jupyter Notebooks
- **Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn, Supervised and Unsupervised Learning
- **Other:** Git, Linux, Microsoft Office Suite
- **Certifications:** Machine Learning Specialisation by Andrew Ng: Supervised and Unsupervised Learning
- **Coding Profiles:** Solved 34 Problems on **LeetCode** (<https://leetcode.com/u/dhruvvarshney/>) and 73 problems on **Codeforces** (<https://codeforces.com/profile/kingdhruv>), with a contest rating of 903.

Extra-Curricular Activities & Achievements

- **Click Kgp Photographer:** A member of the Photography Society of IIT Kharagpur, actively engaged in documenting and showcasing life at campus.
- **Green Belt in Kyokushin Karate:** This accomplishment showcases discipline, diligence, and devotion to physical health.
- **Badminton:** NSO sport and hobby, a sign of my passion for the sport and healthy living.
- **NCSC National Level Selection (Twice):** For contribution towards science communication and research beyond academics.
- **A1 and A2 German Language Exams:** Language proficiency in German certified, evidencing my commitment to multilingualism.
- Competed in various Creative Writing, Quiz and Sports competitions in School and Institute.